

Abstract

Exercises from "Arrows, Structures and Functors: Categorical Imperative"
by Arbib and Manes.

1.1

Exercise 1 Prove that if $f : A \rightarrow B$ is one-to-one then f is a split monomorphism, i.e. that there exists $g : B \rightarrow A$ with $g \circ f = id_A$.

f is injective (one-to-one) if $f(a_1) \neq f(a_2) \forall a \in A, a_1 \neq a_2$. For any $a \in A$ we would have

$$g(f(a)) = a'$$

but since f is injective

$$g(f(a)) = a' = a$$

such that $g \circ f = id_A$.

Exercise 2 Prove that for every function $f : A \rightarrow B$ with A non-empty there exists a "choice function" $g : B \rightarrow A$ such that $f \circ g \circ f = f$. In axiomatic set theory this is an acceptable form of the "axiom of choice".

In such case, $(g \circ f)(a)$ will always give a preimage of f such that

$$(f \circ g \circ f)(a) = f(a)$$

Exercise 3 Let A be any set and let $f : \emptyset \rightarrow A$ be the unique function "inclusion of the empty set". [Note: recall that a function $A \rightarrow B$ is any subset f of $A \times B$ satisfying "given $a \in A$ there exists a unique pair in f with first coordinate a ."] Prove that f is a monomorphism.

First of all, it should be noted that there's only one function f which maps $\emptyset \rightarrow A$, as a subset of $\emptyset \times A = \emptyset$, since the empty set $\emptyset$ has no elements to map from. In order to prove that f is a monomorphism, let's assume two functions

$$g, h : \emptyset \rightarrow \emptyset$$

in order to verify that

$$f \circ g = f \circ h \implies g = h$$

We are defining g and h in this manner, because we can't define a function from a non-empty set to an empty set (e.g. $A \rightarrow \emptyset$, as we can't assign any elements of $\emptyset$ (there are none) to those of A). As such

$$f \circ g = f \circ h : \emptyset \rightarrow A$$

implies necessarily that $g = h$.

Exercise 4 Let 2^A denote the power set of all subsets of A . Given $f : A \rightarrow B$ define $2^f : 2^B \rightarrow 2^A$ by $2^f(S) = f^{-1}(S) = \{a \in A \mid f(a) \in S\}$. Prove that f is onto iff 2^f is one-to-one.

Assuming 2^f is injective (one-to-one), with $2^f(b_1) \neq 2^f(b_2) \forall b \in 2^B, b_1 \neq b_2$, then f^{-1} will also be injective as $2^f(S) = f^{-1}(S)$. If f^{-1} is injective then this means f will be surjective (onto) as every element in codomain B is mapped by at least one element in the domain A .

If 2^f isn't injective then f^{-1} isn't injective, which means $\exists b \in B$ such that $f^{-1}(b_1) = f^{-1}(b_2)$, $b_1 \neq b_2$. This would mean $\exists a \in A$ such that $f(a) = b_1 = b_2$, which doesn't allow f to be properly defined as a function, let alone as an onto one.

Exercise 5 Using the definitions of Exercise 4, prove that if (e, m) is an epi-mono factorization of f then $(2^m, 2^e)$ is an epi-mono factorization of 2^f .

If the pair (e, m) is an epi-mono factorization of $f : A \rightarrow B$, we have $f = m \circ e$, e being an epimorphism $e : A \rightarrow f(A)$ and m being a monomorphism $m : f(A) \rightarrow B$. We can presumably prove this by assuring that the following diagram commutes

$$\begin{array}{ccc} A & \xrightarrow{e} & f(A) \\ 2^e \uparrow & \nearrow \phi & \downarrow m \\ 2^{f(A)} & \xleftarrow{2^m} & B \end{array}$$

Or we can see that if e is an epimorphism, then 2^e will be monomorphic, and likewise if m is a monomorphism, then 2^m will be epimorphic. Thus the pair $(2^m, 2^e)$ will be an epi-mono factorization of 2^f , such that there's an isomorphism ϕ which makes the corresponding diagram commute.